Research Statement

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My research sits at the intersection of algebra and combinatorics, and I am especially drawn to matroid theory and the algebraic structures that shape combinatorial models. I enjoy exploring how algebraic and geometric ideas reveal the underlying structure of combinatorial objects, and how combinatorial viewpoints can, in turn, guide algebraic thinking. I hope to contribute to a more unified understanding that connects algebraic combinatorics, matroid theory, and related areas like tropical geometry and group theory.

What motivates me most is the way matroids capture the idea of independence across so many settings linear algebra, graphs, and field extensions. They offer a language that brings different parts of mathematics together. Many classical problems become clearer when viewed through this lens, whether we are thinking about representability, characteristic sets, or structural dualities. My work aims to uncover and formalize these connections by developing new algebraic tools and studying how algebraic invariants interact with combinatorial structure.

My current research is centered on matroid theory, especially on questions about representations and characteristic sets [CV24]. I have been studying how algebraic data can be used to understand representable matroids, and I have identified new pairs of linear and algebraic characteristic sets. These findings help clarify how algebraic propertiessuch as those coming from field extensions or polynomial relations shape representability and structural behavior in matroids.

This work naturally leads into broader questions in algebraic combinatorics. I am interested in how invariants, polynomial ideals, and geometric or tropical interpretations can reveal deeper combinatorial patterns. In particular, I am excited about exploring how algebraic and tropical varieties might encode matroidal information in ways that connect geometry and combinatorics.

Alongside this, I have also worked on more applied problems in discrete optimization and model selection. One recent project involves determining the optimal polynomial degree for integral solution methods like multiple shooting in dynamical systems. Even though this work is more applied, it is guided by the same algebraic and combinatorial intuitionusing structure to make models more efficient, interpretable, and mathematically meaningful.

Future Directions.

The next step in my research is to continue developing the theory of flock characteristic sets, an invariant that arises from Frobenius flocks and captures deep algebraic information about a matroid. Flock characteristic sets reflect how a matroid behaves under Frobenius twisting and relate closely to questions of representability over fields of positive characteristics. My primary focus lies in exploring

the potential combinations of linear, algebraic, and flock characteristic sets. In other words, given three subsets of $\mathbb{P} \cup \{0\}$, satisfying the constraints on the linear, algebraic, and Frobenius flock characteristic sets, does there exist a matroid with the corresponding characteristic sets? My longterm goal is to build a richer algebraic viewpoint on matroids through representations and characteristic sets. While my work is rooted in matroid theory, I am deeply interested in the broader connections between algebra and combinatorics and look forward to developing these ideas further.

Matroids with Cofinite Flock Characteristic Sets.

One question I am excited to pursue is whether there are matroids whose flock characteristic sets are cofinite. Since Frobenius flocks are closely tied to linear representations, I believe the structure of linear characteristic sets can help guide the search for such examples and clarify when cofinite occurs.

Exploring Other Finite Flock Characteristic Sets.

Another direction is to understand what other finite flock characteristic sets can appear. I have shown several examples already, but many possibilities remain open. I hope to identify the algebraic features that lead to finiteness and to determine which matroids exhibit these patterns.

Applied Directions in Optimization and Dynamical Systems.

Recently, I came across an exciting research area in optimization and dynamical Systems called Sparse Identification of Nonlinear Dynamical Systems (SINDy) [BPK16]. SINDy is an effective data driven technique for determining the underlying dynamics of complicated systems. It is especially helpful in disciplines like engineering, biology, and physics where nonlinear differential equations are frequently used to explain systems. SINDy looks for the most straightforward mathematical model that can be created from data to represent the observed behavior of a system.

Discrete Optimization for Choosing Polynomial Degrees.

My current work is to study how discrete optimization can be used to determine the most appropriate polynomial degree in integral solution methods. The goal is to develop principled, combinatorial criteria that improve on existing approaches.

This combination of algebraic insight and algorithmic problem solving reflects my broader interest in the connections between algebra, combinatorics, and theoretical computer science.

References

[BPK16] Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz, *Discovering governing equations from data by sparse identification of nonlinear dynamical systems*, Proceedings of the National Academy of Sciences **113** (2016), no. 15, 3932–3937.

[CV24] Dustin Cartwright and Dony Varghese, *Characteristic sets of matroids*, European Journal of Combinatorics **118** (2024), 103939.